

a Study region and spring migration direction field

- Onshore wind farm (n = 4,191)
- Offshore farm, VPTS radar (n = 29)
- Offshore farm, Bauer grid (n = 26)
- Weather radar station (n = 6)

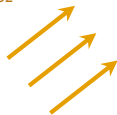
Spring migration direction concentration



Mechanism chain (b → f): array orientation, not siting alone, sets exposure

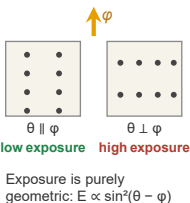
b Migration field ϕ

$\phi \approx 52^\circ$

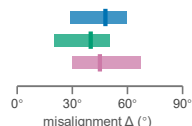


2,025 grid cells; mean spring bearing 52° , direction concentration 0.89

c Array orientation θ

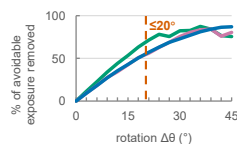


d Energy optimum ≠ ecological optimum



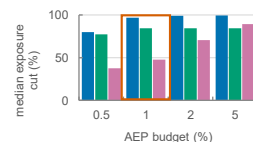
Median Δ = 48° onshore, 40° VPTS, 45° Bauer (top to bottom); bars = IQR

e A small rotation captures most of the gain



$\Delta\theta_{50} = 17^\circ$ onshore; a $\leq 20^\circ$ rotation already captures 55% of the avoidable exposure

f A $\leq 1\%$ AEP budget buys nearly all of it



At a 1% AEP budget the median exposure cut is 97% onshore, 84% VPTS, 48% Bauer